

Course 6009

Semester 6

Project

Shared handbook

Major Project - Phase II

Complete, verify, document and present the major project using the approved Phase-I baseline, controlled changes and evidence-based evaluation.

Course code	6009
Revision	REV2026
Semester	Semester 6
Type	Project
Catalogue scope	42 catalogue programme assignments

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 6009 — Major Project - Phase II**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

Official SITTTTR Revision 2026 syllabus reference. This page is a study handbook, not a substitute for the latest official notification or examination instruction.

COURSE ORIENTATION

How to use this handbook

This page is a structured learning aid for the shared catalogue record. Study the concept, complete the applied activity, retain the evidence, and compare the result with the latest approved programme instructions. Where the official syllabus differs, the official record takes precedence.

Source and verification note: The course identity, code, semester and subject type are taken from the tracked POLY PMNA REV2026 catalogue, which indexes the official SITTTTR scheme. The official course-content reference is SITTTTR course 6009; the scheme index is SITTTTR REV2026 diploma syllabus. The direct subject-content page was not machine-readable or returned an unavailable-file response during preparation, so module headings and explanatory teaching material on this page are an original study-handbook structure, not claimed verbatim SITTTTR wording. Confirm current hours, marks, credits, outcomes and assessment against the latest official programme record before examination use.

Learning outcomes

- Explain the central concepts and vocabulary of the subject using clear technical language.
- Apply a controlled project workflow to a diploma-level problem with scope, milestones and acceptance evidence.
- Create a proposal, plan, design or final evidence pack that a supervisor can review.
- Identify assumptions, limitations and common errors, then improve the work using feedback.

Shared-record context

This code is assigned to 42 catalogue programme assignments in the tracked catalogue. The page therefore focuses on common learning practice and avoids claiming

programme-specific technical details that were not verified.

Study rule: Do not treat the author-created examples as official examination questions or official mark distribution.

STUDY MAP

Modules and evidence

Author-created study map; verify official hours and marks

Module	Heading	Purpose	Evidence to retain
1	Phase-II baseline and execution plan	Start implementation from the approved Phase-I scope and make any change visible and reviewable.	Create a Phase-II execution board with milestones, evidence and risk responses.
2	Implementation and integration	Develop or execute the approved solution incrementally and integrate its components responsibly.	Record one integration test with setup, expected result, actual result and corrective action.
3	Verification, validation and evaluation	Demonstrate that the project satisfies requirements and explain limitations honestly.	Complete a requirement-to-evidence matrix and write an evaluation paragraph for each unmet target.
4	Report, demonstration and presentation	Communicate the completed work to technical and non-technical reviewers with clear evidence.	Prepare a ten-slide presentation outline and a demonstration checklist.
5	Closure, handover and reflection	Close the project professionally and leave an understandable, maintainable record.	Complete a closure report that links every project objective to evidence and a future action.

MODULE 1

Phase-II baseline and execution plan

Purpose-led study

Apply + reflect

Start implementation from the approved Phase-I scope and make any change visible and reviewable.

Core topics–

- Baseline review and open issues
- Implementation milestones
- Roles, resources and supervision

- Change requests and decision log

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Create a Phase-II execution board with milestones, evidence and risk responses.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections–

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check—

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

Malayalam support: ു ഐക്യപരമായ പാഠങ്ങൾ ഉൾക്കൊള്ളുന്ന English technical terms-
 ുൾക്കായി തയ്യാറാക്കിയിരിക്കുന്നു. ഈ പാഠങ്ങൾക്ക് പദസമുച്ചാരം, formula, diagram,
 units, definitions ുൾക്കായി ഉൾക്കൊള്ളിക്കപ്പെട്ടിരിക്കുന്നു.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

MODULE 2

Implementation and integration

Purpose-led study

Apply + reflect

Develop or execute the approved solution incrementally and integrate its components responsibly.

Core topics–

- Incremental development and configuration
- Interfaces and integration points
- Version control and documentation
- Safety, ethics and data handling

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Record one integration test with setup, expected result, actual result and corrective action.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections–

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.

- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check–

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 3

Verification, validation and evaluation

Purpose-led study

Apply + reflect

Demonstrate that the project satisfies requirements and explain limitations honestly.

Core topics–

- Test plan and traceability
- Functional, performance, usability and safety checks
- Results, defects and retesting
- Evaluation against objectives and acceptance criteria

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Complete a requirement-to-evidence matrix and write an evaluation paragraph for each unmet target.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections–

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check–

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 4

Report, demonstration and presentation

Purpose-led study

Apply + reflect

Communicate the completed work to technical and non-technical reviewers with clear evidence.

Core topics–

- Final report chapters and appendices
- Figures, tables, citations and reproducibility
- Demonstration script and contingency plan
- Presentation, viva and questions

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Prepare a ten-slide presentation outline and a demonstration checklist.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections–

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check–

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 5

Closure, handover and reflection

Purpose-led study

Apply + reflect

Close the project professionally and leave an understandable, maintainable record.

Core topics–

- Final artefact and source/archive package

- User or maintainer handover
- Individual contribution evidence
- Limitations, lessons learned and future work

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Complete a closure report that links every project objective to evidence and a future action.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections–

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check–

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

Evidence, revision and quality gates

Before submission or examination

1. Confirm the current SITTTTR/SBTE title, hours, credits, outcomes and assessment.
2. Define the problem or prompt and state assumptions.
3. Show the method, calculation, algorithm, project record or communication structure.
4. Attach test, source, supervisor, workplace or reflection evidence as applicable.
5. Review clarity, safety, ethics, citations, originality and accessibility.

Revision checklist

- Can I define the main terms without copying?
- Can I apply the method to a new situation?
- Can I explain one common error and its correction?
- Can I show evidence for each claimed outcome?
- Can I state the limitations and the next improvement?

Evidence record

Subject: 6009 – Major Project - Phase II

Task or question:

Assumption or requirement:

Method / design / procedure:

Result or artefact:

Test / source / supervisor evidence:

Reflection and next action:

SELF-CHECK AND EXAMINATION PRACTICE

Question bank

Recall question

Define two important terms from Major Project - Phase II and distinguish them.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Explain question

Explain why the method in this handbook matters for a diploma student.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Apply question

Apply one module method to a realistic technical, project or workplace situation.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Evaluate question

Identify a weak approach, state the evidence needed, and propose a defensible improvement.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Create question

Design a small artefact, plan, test, report section or presentation that demonstrates the subject outcomes.

Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

REFERENCES AND GLOSSARY

Reference trail

1. Official SITTTT REV2026 diploma syllabus catalogue.
2. Official SITTTT course-content reference for code 6009.
3. POLY PMNA Study Hub, for the current revision index and related lesson pages.

Glossary

Terms used in this handbook

Term	Working meaning
Outcome	A capability the learner should demonstrate, not just a topic read.
Evidence	A record, artefact, result, source, test or review that supports a claim.
Acceptance criterion	A condition used to decide whether a requirement or deliverable is satisfactory.
Reflection	A brief evidence-based account of what worked, what did not and what should change.

Prepared as a POLY PMNA study handbook. Always follow the latest official SITTTTR/SBTE instructions for the programme and examination session.